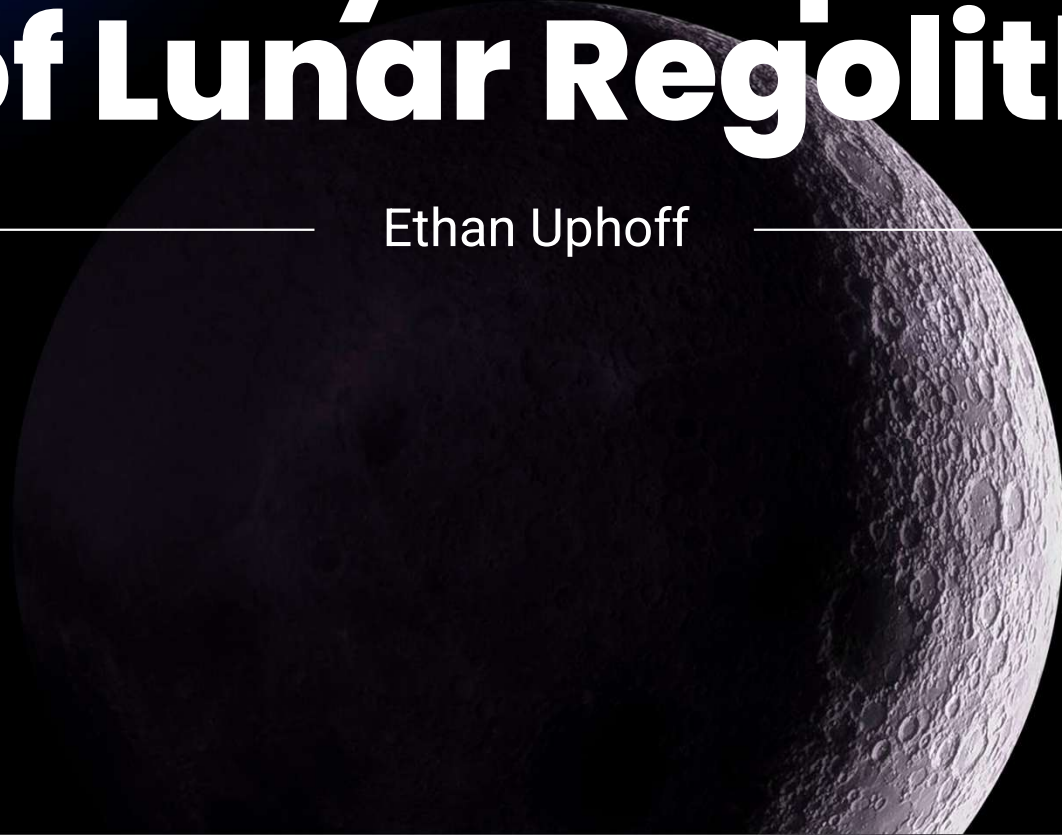


Vibratory Compaction of Lunar Regolith

Ethan Uphoff



Presentation Overview

01

**Problem and
Motivation**

02

**Background &
Existing Methods**

03

**Concept
Development &
Trade Study**

04

**Key Innovation:
Vibratory
Compaction**

05

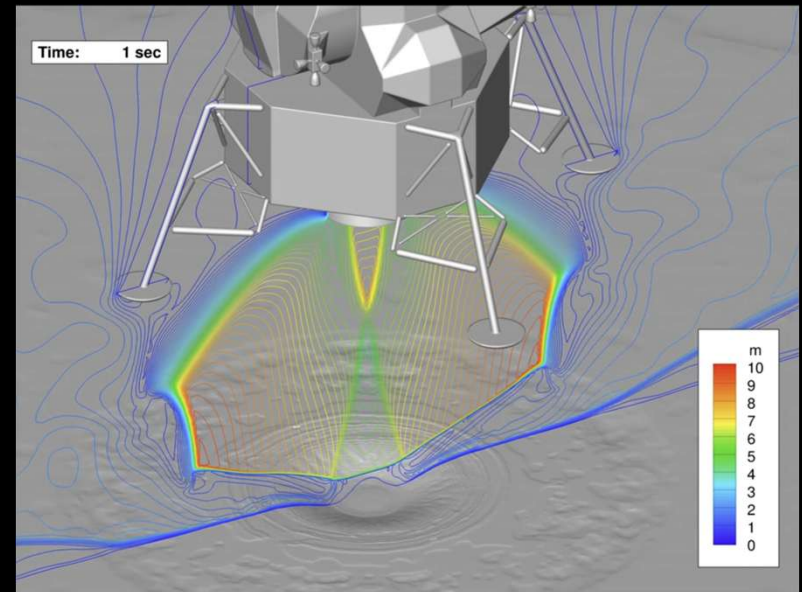
**Design &
Experimental
Validation**

06

**Results and
Conclusions**

Background

- Lunar regolith is loose, granular, and weakly compacted due to low gravity
- Lander plume ejecta can damage equipment and erode the surface
- Surface compaction is required for safe landing pads, roads, and infrastructure



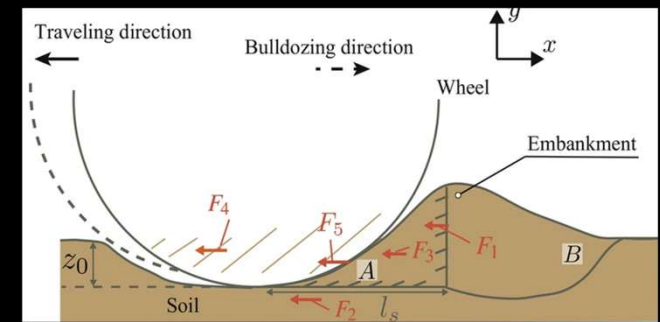
Problem Statement

The success of sustained lunar missions depends not only on reaching the Moon, but on preparing its surface. However, loose and unstable regolith leads to hazardous dust plumes, poor traction, and unsafe load-bearing conditions, while existing rover systems lack the capability to actively compact and stabilize the terrain.



Heritage of Compaction

- Bekker Terramechanics
- Compaction on Earth
- NASA's PACT and STOMP



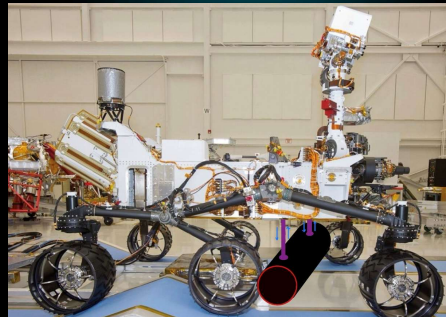
3 Concepts Were Investigated

Concept 1



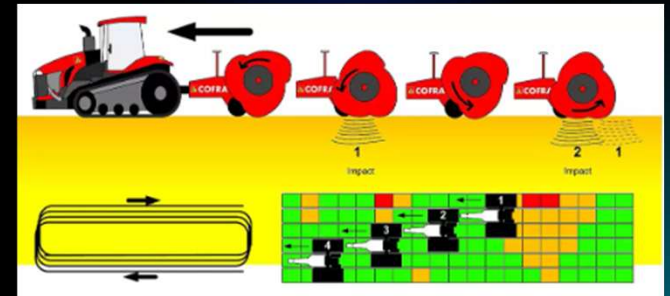
Vibratory Roller Wheel

Concept 2



Adjustable Vibratory Roller

Concept 3



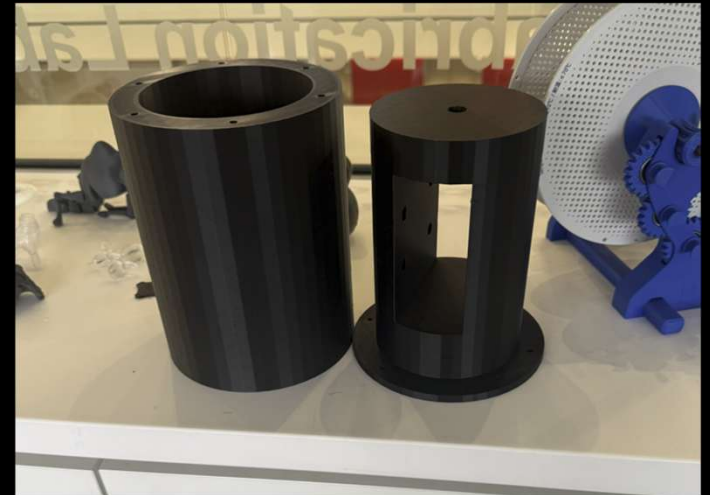
CRC Roller Trailer

Weighted Decision Matrix

		Concept 1 (Wheel)		Concept 2 (Adjustable)		Concept 3 (CRC)	
Parameter	Weight	Score	Rating	Score	Rating	Score	Rating
Compaction	30	7	210	9	270	8	240
Compaction Efficiency	30	8	240	8	240	6	180
Slip Resistance	30	6	180	10	300	8	240
Mobility	20	10	200	10	200	7	140
Energy Efficiency	15	6	90	8	120	10	150
Moon Resistance	15	8	120	8	120	5	75
Mass / Size	10	10	100	8	80	5	50
Risk / Longevity	10	6	60	8	80	9	90
Autonomy	10	10	100	10	100	10	100
Difficulty / Complexity	10	8	80	9	90	7	70
Total	-----	-----	1380	-----	1600	-----	1335

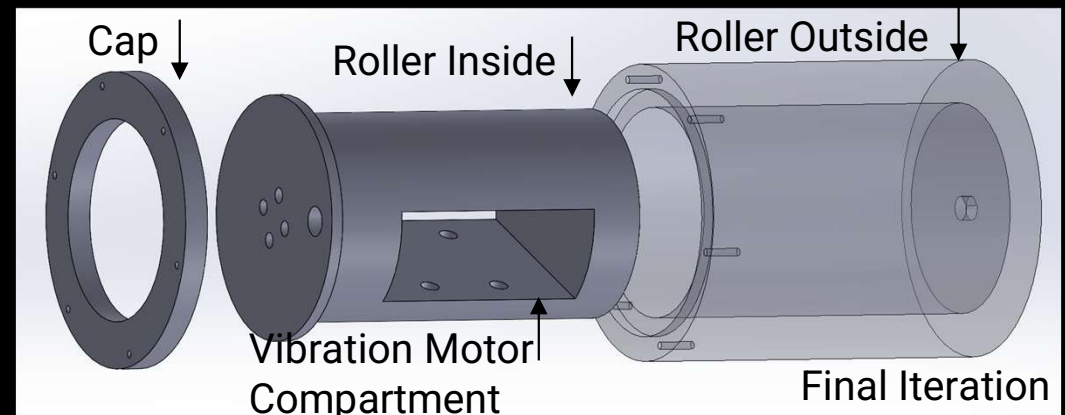
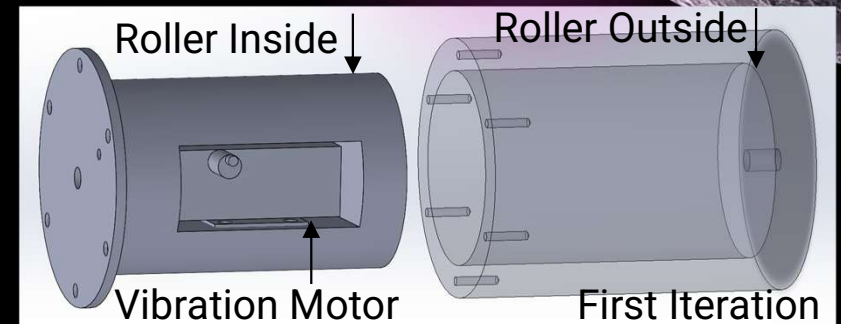
Key Innovations

- The vibratory roller can be implemented in wheels or adaptable on a rover
- A vibratory mechanism has never been put in wheels before for compaction
- Enables continuous, in-situ regolith compaction during rover traversal



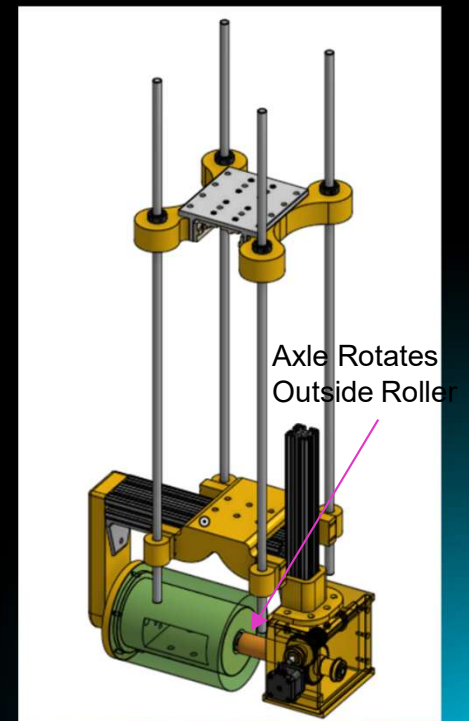
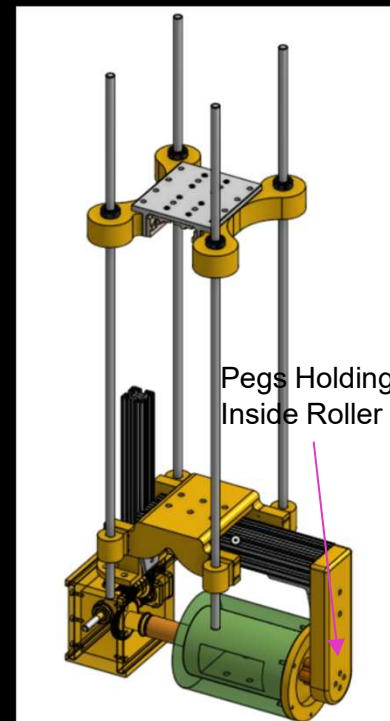
Design Implementation

- Three total iterations
- Internal vibratory motor integrated within roller housing
- Final iteration was modified to be tested on RIDER Jr.



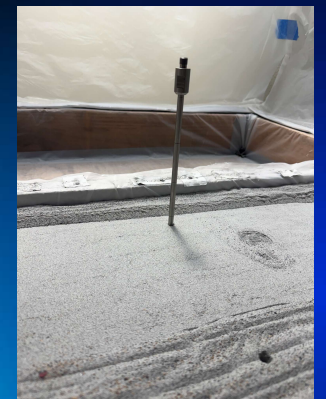
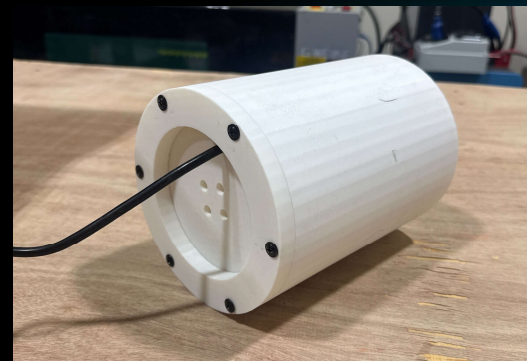
RIDER Jr. Test Bed

- Position of roller in RIDER Jr.
- Constant speed across regolith bed
- Supported by metal frame



Experimental Setup

- Two runs to compare compaction:
 - Vibration motor off
 - Vibration motor on (different freq.)
- Penetration force and depth measured
- Density and type of regolith, speed, RIDER Jr., and number of passes all held constant for each run.



Results

							Measurement (in)
Trial	Vibration Frequency (Hz)	Force B1 (kgf)	Force A1 (kgf)	Depth B1 (in)	Depth A1 (in)		
1	0 Hz	2.53 kgf	2.72 kgf	2.65 in	2.38 in		Cone Diameter 0.4165
2	0 Hz	2.68 kgf	2.97 kgf	2.58 in	2.45 in		Cone Height 0.6715
3	0 Hz	2.36 kgf	3.49 kgf	2.71 in	2.30 in		washer diameter 1.082
4	0 Hz	3.17 kgf	4.48 kgf	2.55 in	2.27 in		
5	0 Hz	3.39 kgf	3.94 kgf	2.56 in	2.26 in		Penetrometer Surface Area (in^2): 0.9195
6	1728 Hz	3.68 kgf	5.37 kgf	2.50 in	2.20 in		
7	1745 Hz	4.64 kgf	4.97 kgf	2.55 in	2.25 in		
8	1820 Hz	4.51 kgf	4.69 kgf	2.52 in	2.18 in		
9	855 Hz	4.37 kgf	4.44 kgf	2.49 in	2.25 in		
10	1149 Hz	5.25 kgf	6.21 kgf	2.46 in	2.34 in		
11	2415 Hz	4.98 kgf	4.18 kgf	2.49 in	2.30 in		
12	2805 Hz	5.72 kgf	6.33 kgf	2.57 in	2.28 in		
13	1420 Hz	3.49 kgf	4.38 kgf	2.80 in	2.24 in		
14	3511 Hz	5.85 kgf	6.68 kgf	2.55 in	2.28 in		
15	3195 Hz	4.11 kgf	4.11 kgf	2.41 in	2.36 in		

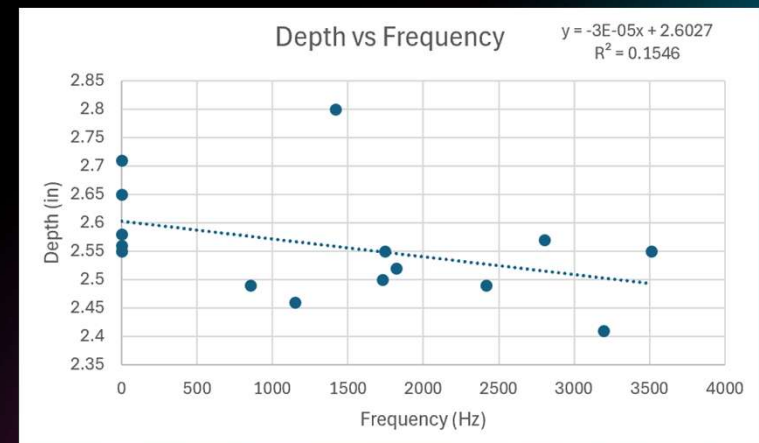
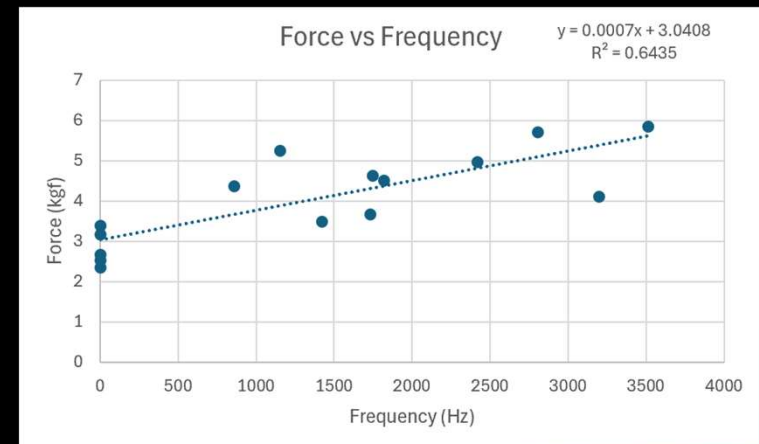
Frequency Range (Hz)	Mean Stress (psi)	Std Dev Stress (psi)	Mean Depth (in)	Std Dev Depth (in)
0 Hz (5)	7.61 psi	0.88 psi	2.471 in	0.157 in
1-2000 Hz (6)	11.20 psi	1.05 psi	2.398 in	0.124 in
2000+ Hz (4)	12.58 psi	0.65 psi	2.405 in	0.076 in

Analysis & Discussion

- A one-way ANOVA test was conducted to evaluate the effect of vibration frequency on penetration force and compaction depth at a 95% confidence level
 - Penetration Force: $p = 0.0002 < 0.05$
 - Depth: $p = 0.18 > 0.05$

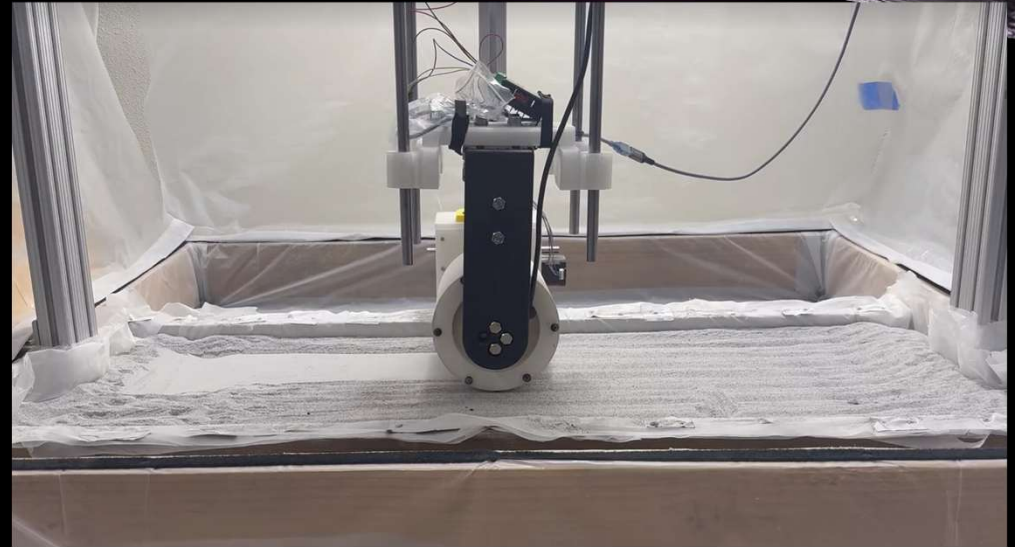
• Linear Correlation for Force and Depth vs Frequency

• Vibration increases penetration resistance, but not compaction depth



Conclusion

- Vibration frequency significantly increases penetration resistance, with stress rising by ~65% from static to high-frequency conditions, confirming a strong effect of dynamic loading.
- Compaction depth showed minimal and statistically insignificant change, indicating that vibration alters resistance to penetration without substantially increasing overall densification.
- Overall results suggest a rate-dependent stiffening response, where higher frequencies increase material resistance, up until a certain unknown point



Future Work

- Test what frequency compacts regolith the best
- Scale up experimental results
- Create prototypes to be placed on actual rovers



References and Acknowledgements

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2. National Aeronautics and Space Administration. (2025). *STOMP: SRR poster on compaction systems*. NASA Technical Reports Server. .

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Questions?

